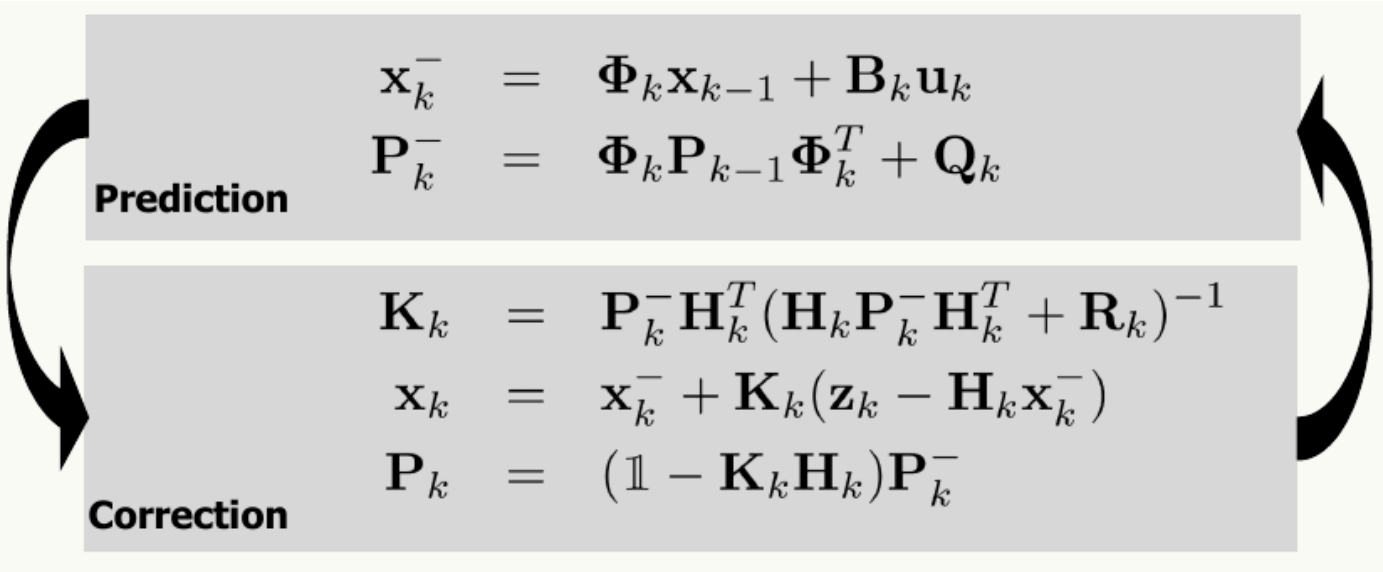
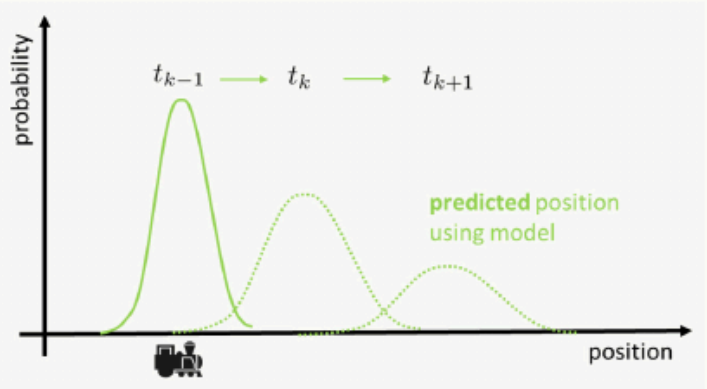
Ms06: Kalman Filter I

Trajectory Estimation

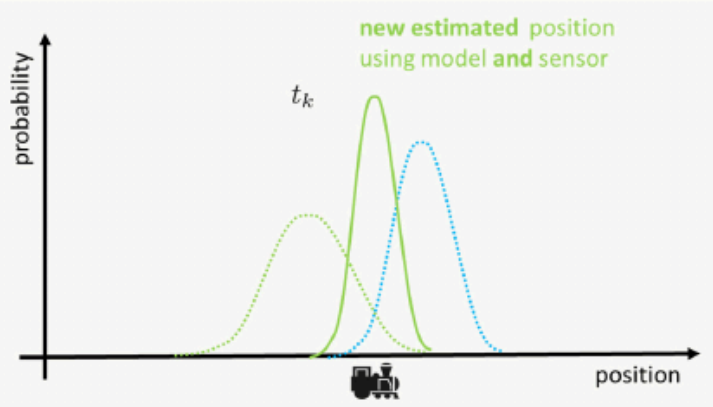
- **Goal**:
  - Estimation of 6 parameters (position and rotation) at each relevant point in time (time of the mapping sensor measurement)
- **Problem**
  - High accuracy via overdetermination (multiple measurements) not possible due to platform motion
- **Solution**:
  - Usage of various partly redundant sensors
  - Usage of knowledge about the system (e.g. motion models)
  - Usage of knowledge about the environment (e.g. maps)
- **Methods**:
  - Sensor fusion, recursive Bayes estimation, Kalman Filter, Bundle adjustment, SLAM

Kalman Filter Algorithm



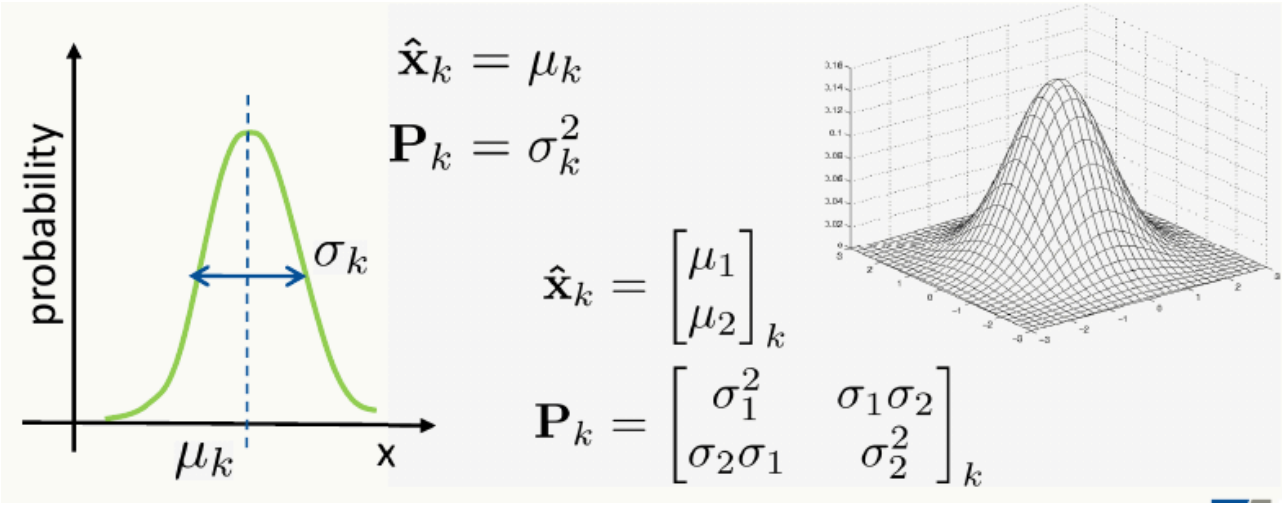


- **Prediction**: use model to predict the state based on the last estimate

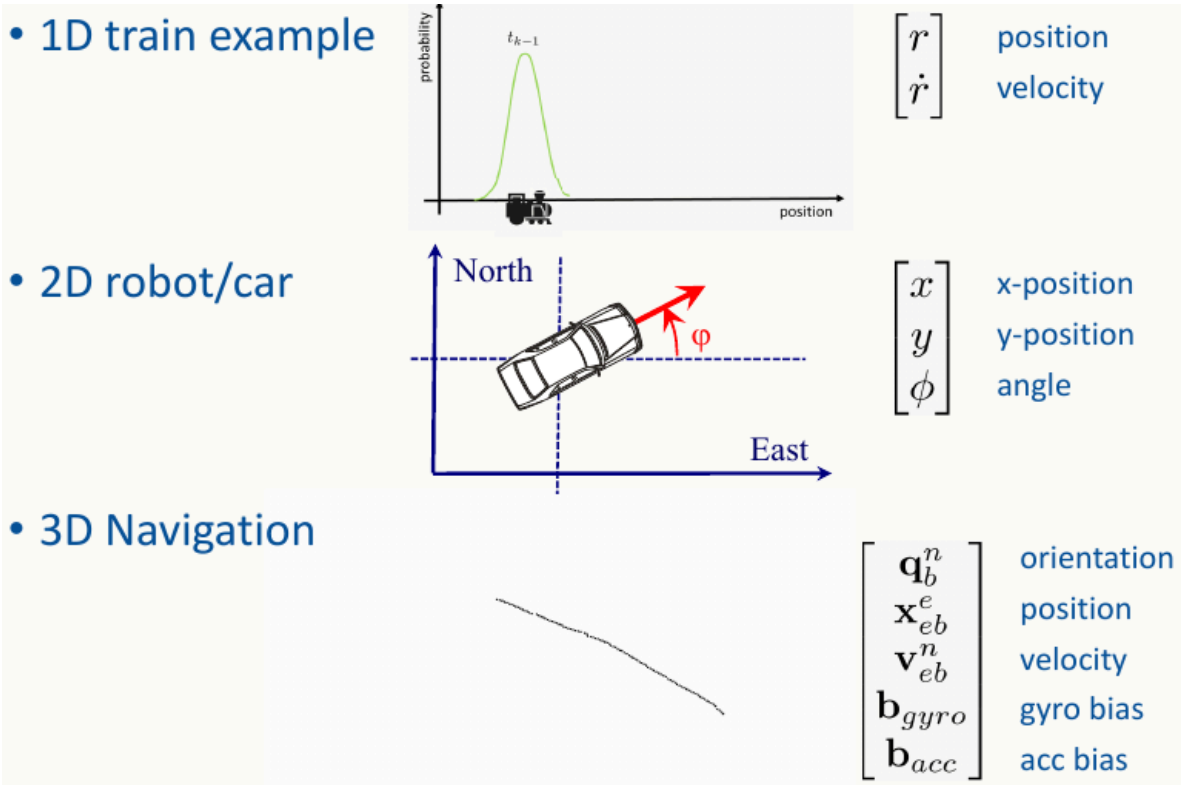


- **Correction:** Combine prediction (model) and sensor observations to update the current estimate

What is needed?

- A way to **represent** the **estimate** (including uncertainty)
  - **State vector** describes the state of the system
  - Probability is a **normal distribution**
  - -> The state vector estimate at a time t_k can be represented using the **mean** $\hat{\boldsymbol{x}}_k$ and the **covariance** $\boldsymbol{P}_k$
  - 

  - Alternative Representation
    - Represent probability distributions using deterministically chosen points (sigma points)
      - -> **Unscented Kalman Filter**
    - Represent probability distributions as set of statistically drawn samples

- -> **Particle Filter**
  - 

- A way to describe the **temporal evolution of the estimate** (including uncertainty)
  - The temporal evolution of the state is described by the **system model** (sometimes called **process model**)

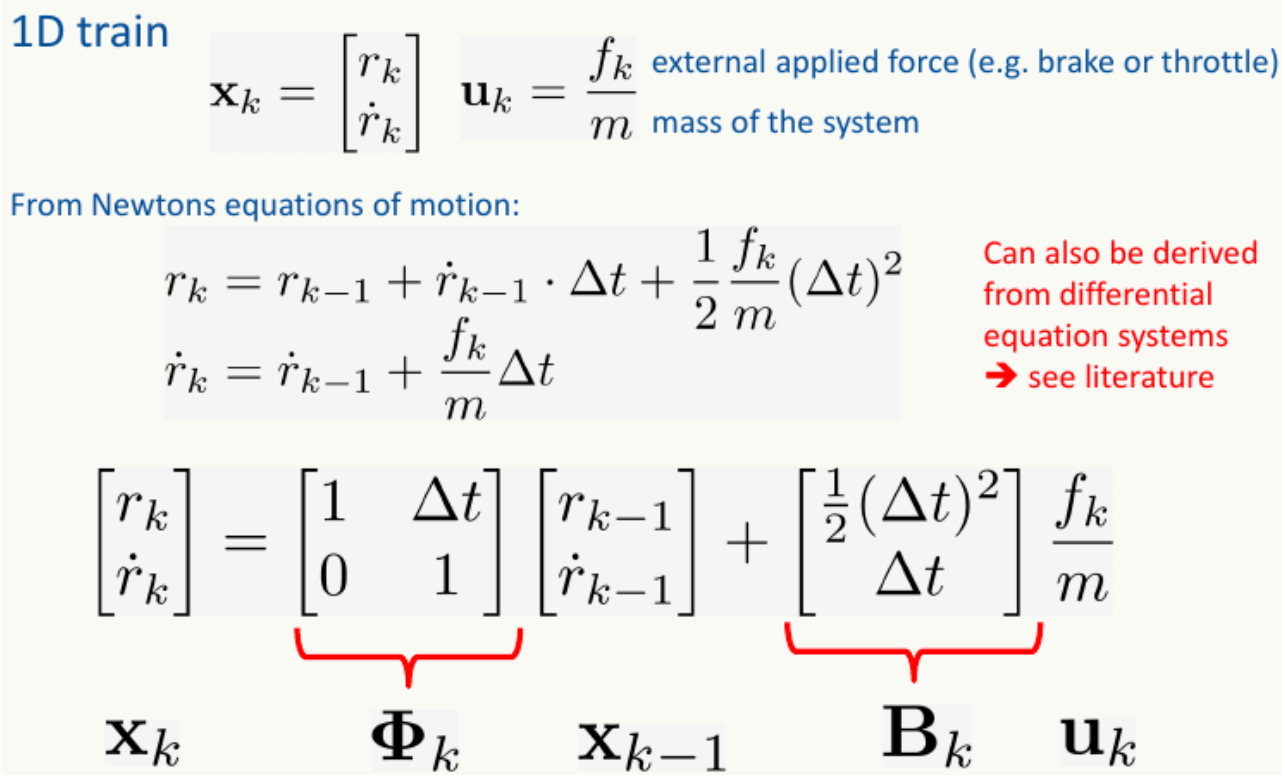
$$\mathbf{x}_k = \mathbf{\Phi}_k \mathbf{x}_{k-1} + \mathbf{B}_k \mathbf{u}_k + \mathbf{w}_k$$

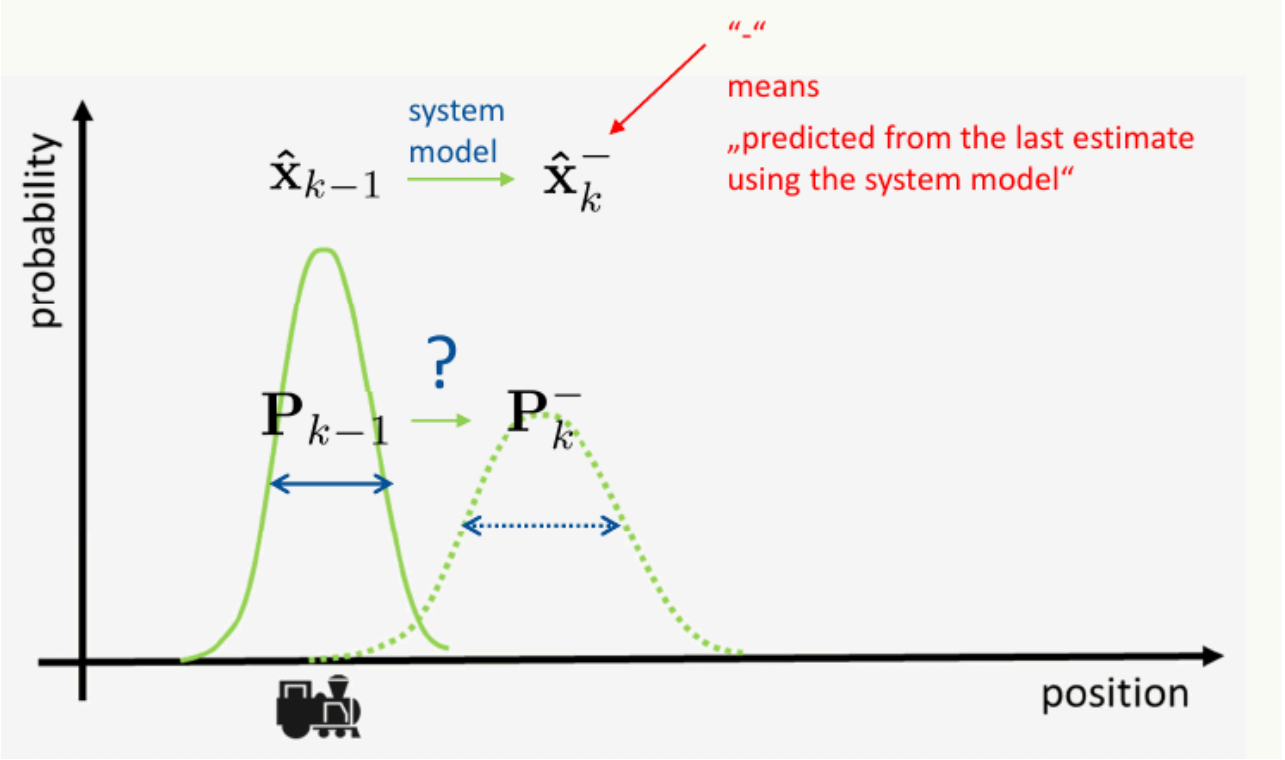
$\mathbf{\Phi}_k$ The **state transition matrix** describes the effect of the old state on the new state

$\mathbf{u}_k$ **control input** vector (e.g. throttle, break, steering)

$\mathbf{B}_k$ The **control input matrix** describes the effect of some control input on the state

$\mathbf{w}_k$ The **process noise** represents all uncertainties of the system model and is assumed to be normally distributed with covariance $\mathbf{Q}_k$

- 

- 

  - The estimate at t_{k-1} is $\hat{\boldsymbol{x}}_{k-1}$ with covariance $\boldsymbol{P}_{k-1}$
  - After predicting $\hat{\boldsymbol{x}}_{\boldsymbol{k}}^-$:
  - $\boldsymbol{P}_k^- = \boldsymbol{\Phi}_k \boldsymbol{P}_{k-1} \boldsymbol{\Phi}_k^T + \boldsymbol{Q}_k$
- A way to describe the **measurement** (including uncertainty)

- The **measurement model** (sometimes also called **observation model**) is describing what the measurement for a given state vector is

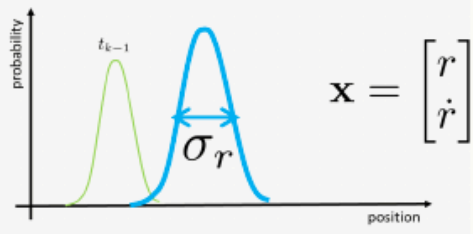
$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k$$

$\mathbf{z}_k$ The **measurement vector** contains the measurements taken at a time t_k

$\mathbf{H}_k$ **The measurement** or **observation matrix** is linking the state to the measurements

$\mathbf{v}_k$ The **measurment noise** represents the uncertainties of the measurement model and is assumed to be normally distributed with covariance $\mathbf{R}_k$

- 1D train model position sensor



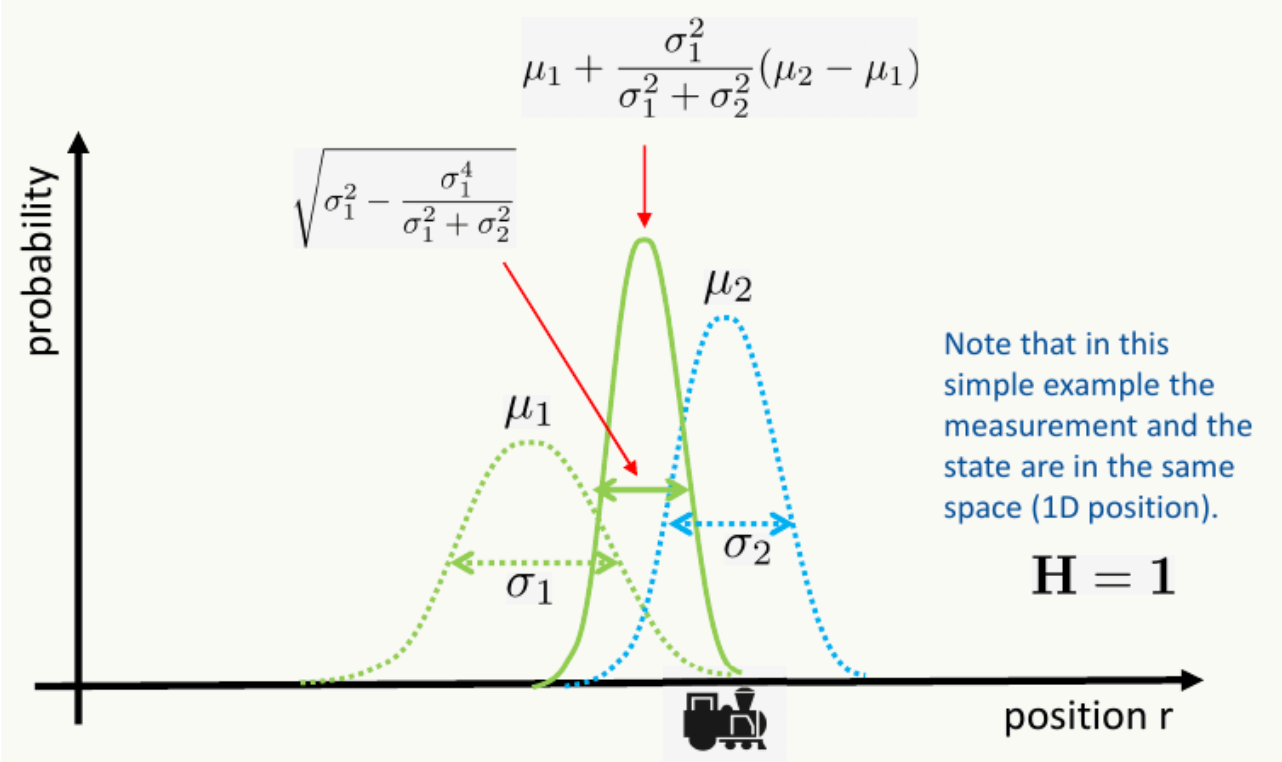
$$\mathbf{x} = \begin{bmatrix} r \\ \dot{r} \end{bmatrix} \qquad z_r = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} r \\ \dot{r} \end{bmatrix} = r \qquad \mathbf{R} = \sigma_r^2$$

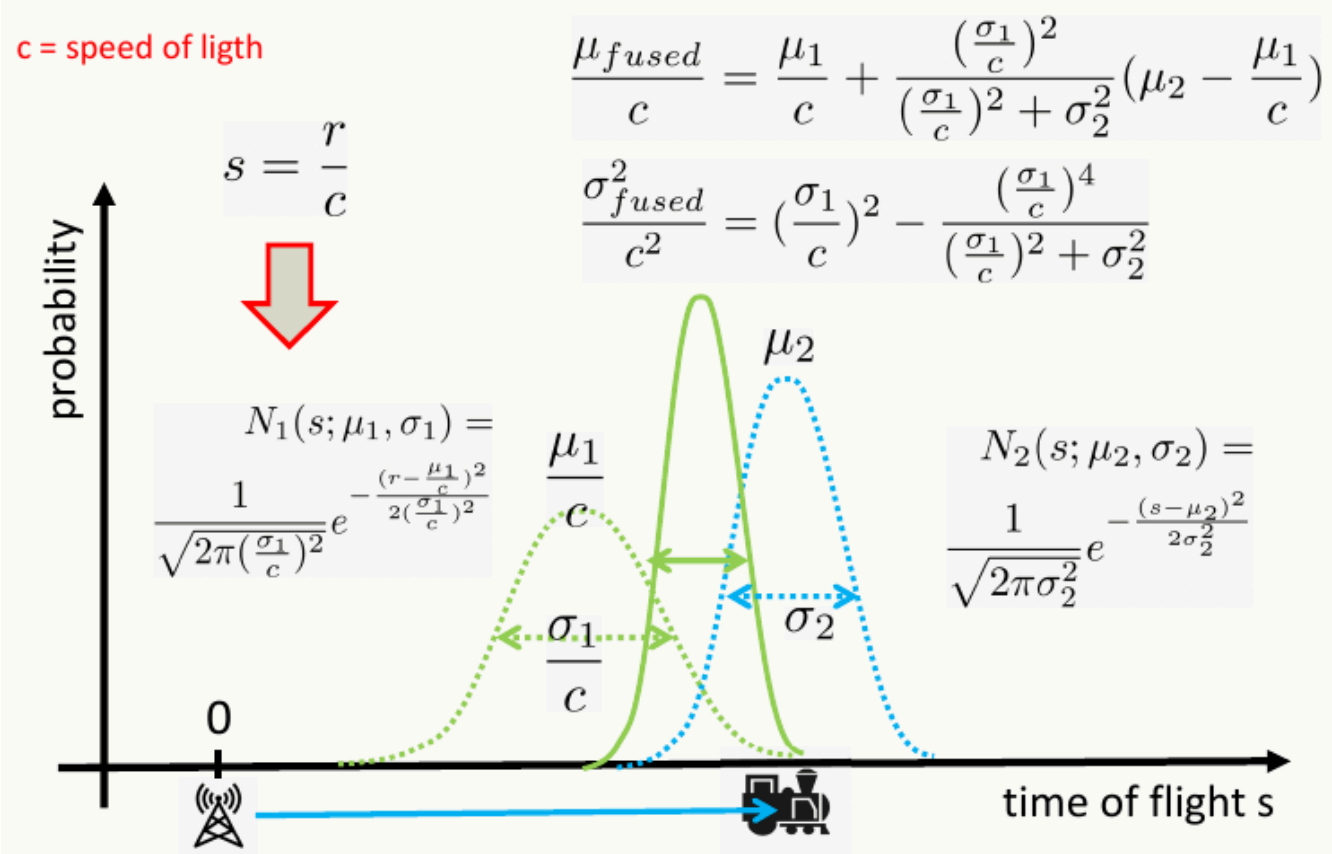
- 3D GPS measurement

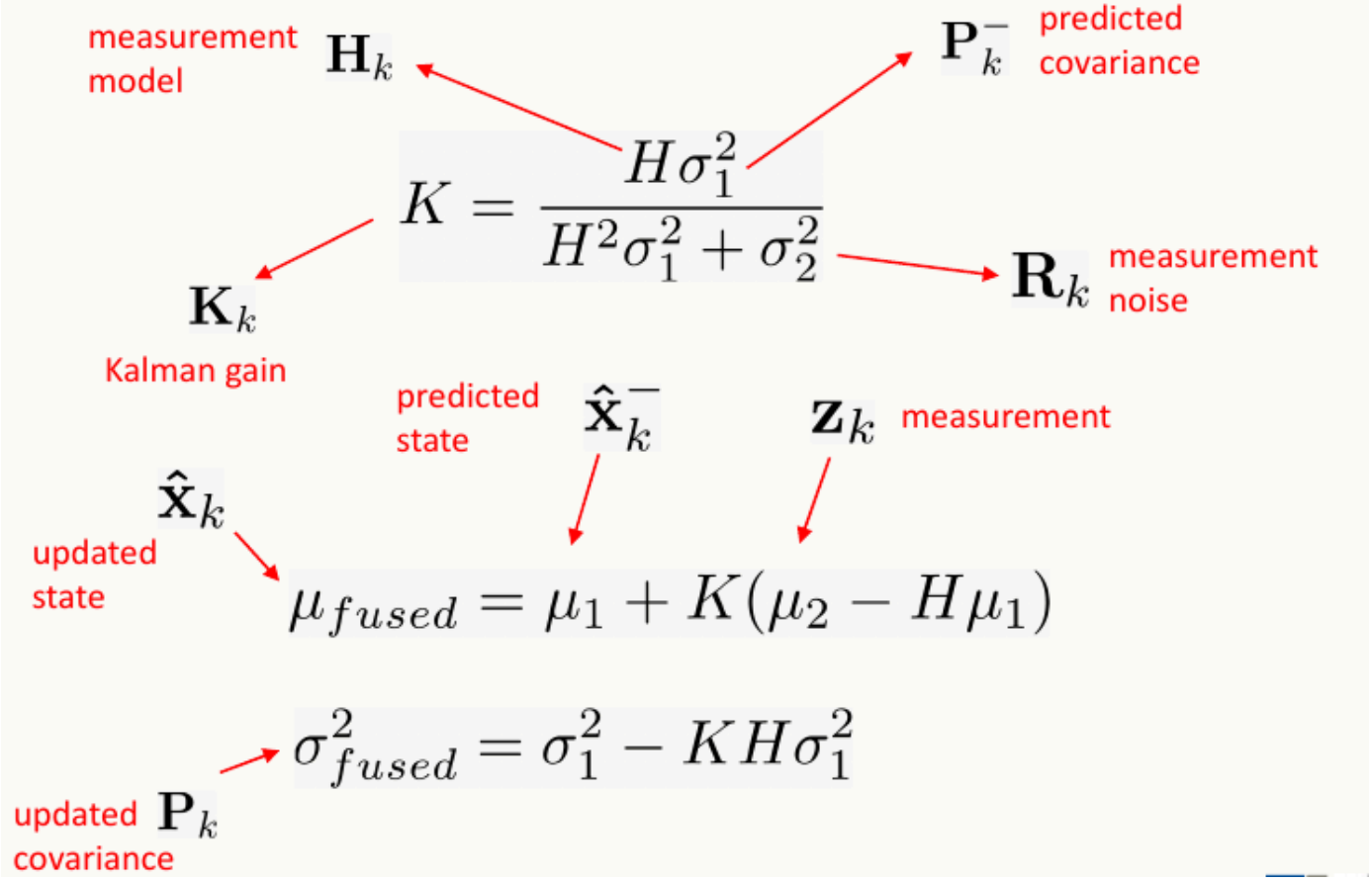
$$\begin{bmatrix} \mathbf{q}_b^n \\ \mathbf{x}_{eb}^e \\ \mathbf{v}_{eb}^n \\ \mathbf{b}_{gyro} \\ \mathbf{b}_{acc} \end{bmatrix} \qquad \mathbf{z}_{gps} = \begin{bmatrix} x_{gps} \\ y_{gps} \\ z_{gps} \end{bmatrix} = \overbrace{\begin{bmatrix} \dots & 1 & 0 & 0 & \dots \\ \dots & 0 & 1 & 0 & \dots \\ \dots & 0 & 0 & 1 & \dots \end{bmatrix}}^{\mathbf{H}} \mathbf{x}$$

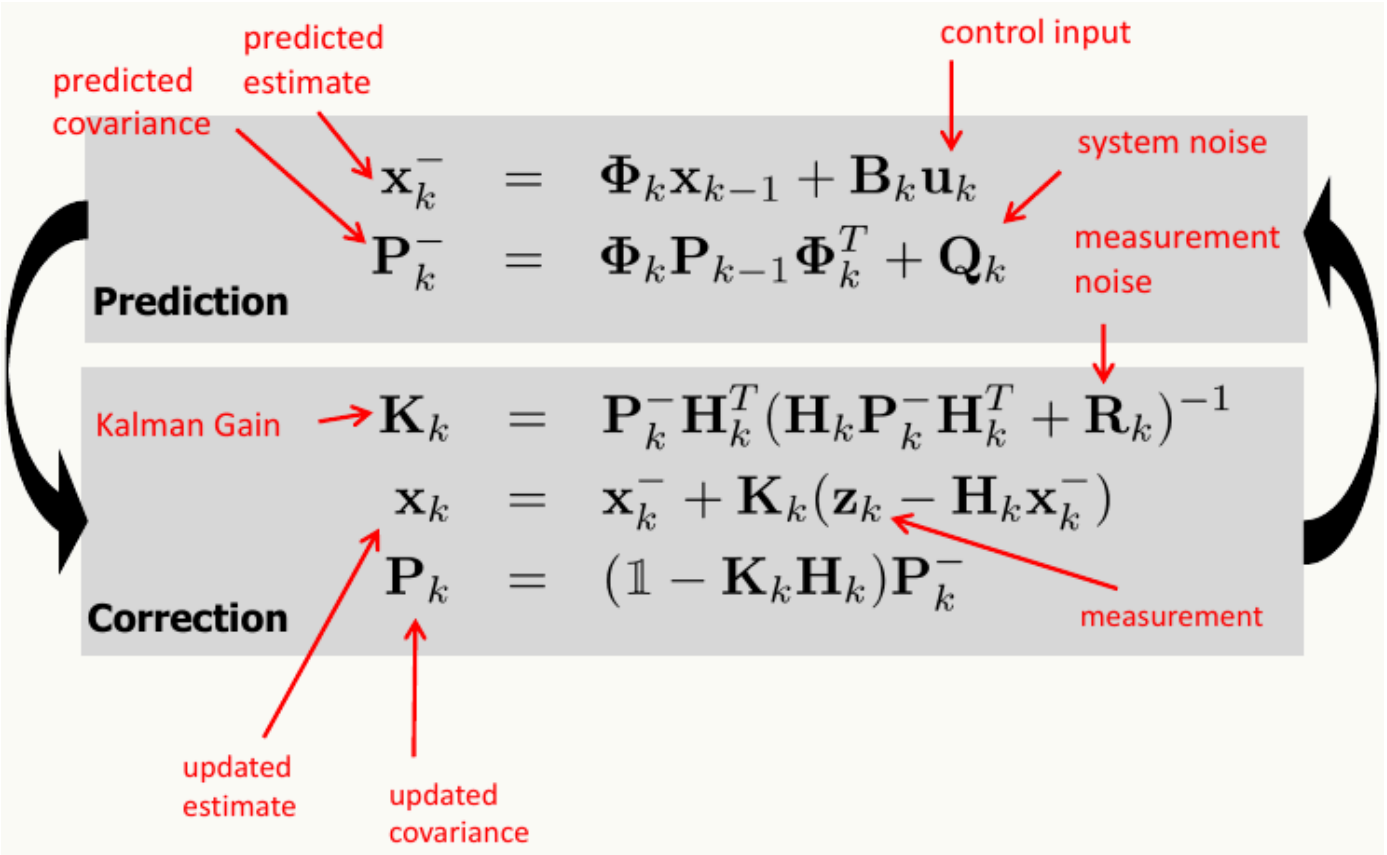
$$\mathbf{R} = \begin{bmatrix} \sigma_{gps,x}^2 & 0 & 0 \\ 0 & \sigma_{gps,y}^2 & 0 \\ 0 & 0 & \sigma_{gps,z}^2 \end{bmatrix}$$

- A rule to **combine** prediction and measurement
  - Weighted Mean of combination (multiplication) of 2 gaussians

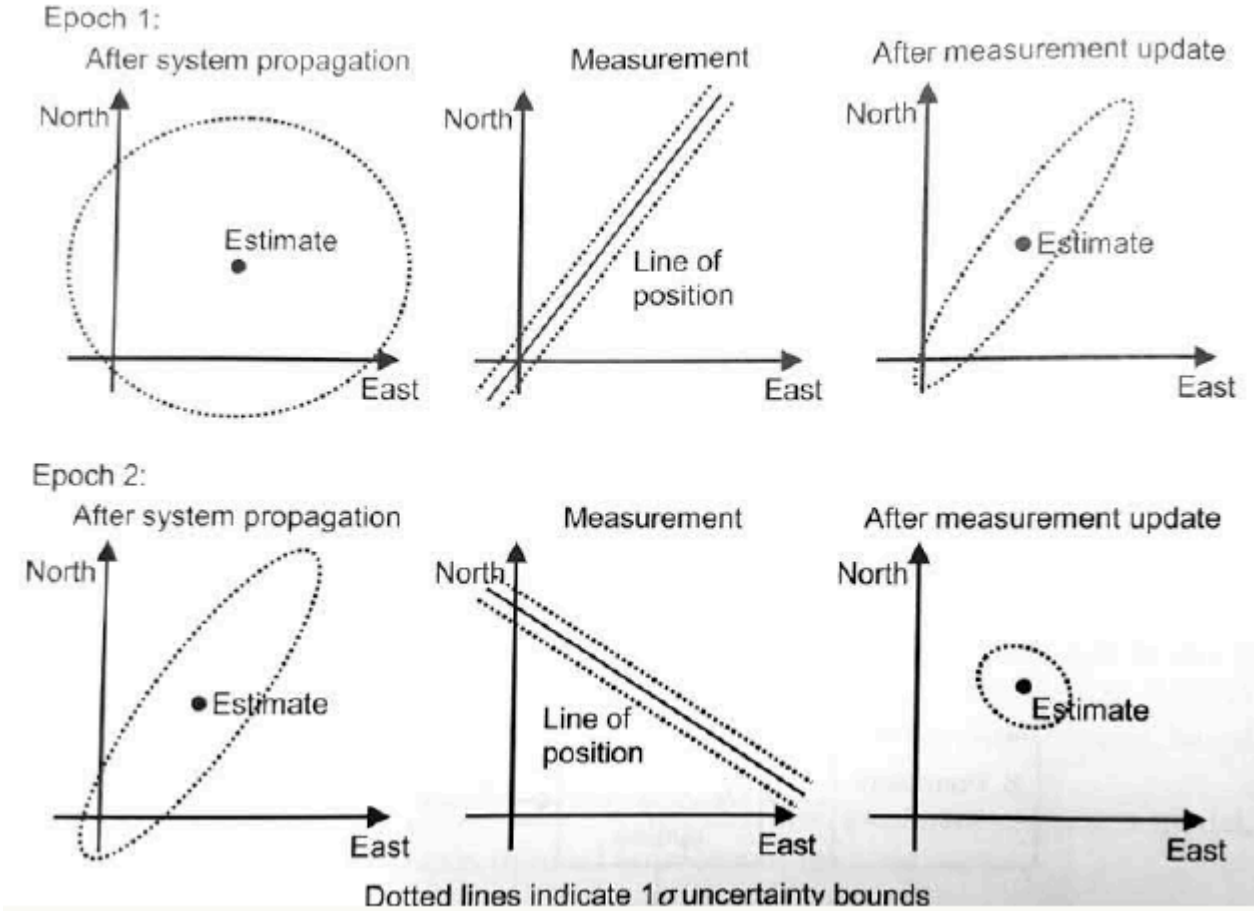
- 
$\mu_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}(\mu_2 - \mu_1)$
$\sqrt{\sigma_1^2 - \frac{\sigma_1^4}{\sigma_1^2 + \sigma_2^2}}$
probability
μ_2
μ_1
σ_2
σ_1
Note that in this simple example the measurement and the state are in the same space (1D position).
$\mathbf{H} = \mathbf{1}$
position r

- 
c = speed of ligth
$\frac{\mu_{fused}}{c} = \frac{\mu_1}{c} + \frac{(\frac{\sigma_1}{c})^2}{(\frac{\sigma_1}{c})^2 + \sigma_2^2}(\mu_2 - \frac{\mu_1}{c})$
$s = \frac{r}{c}$
$\frac{\sigma_{fused}^2}{c^2} = (\frac{\sigma_1}{c})^2 - \frac{(\frac{\sigma_1}{c})^4}{(\frac{\sigma_1}{c})^2 + \sigma_2^2}$
probability
μ_2
$N_1(s; \mu_1, \sigma_1) = \frac{1}{\sqrt{2\pi(\frac{\sigma_1}{c})^2}} e^{-\frac{(r-\frac{\mu_1}{c})^2}{2(\frac{\sigma_1}{c})^2}}$
$\frac{\mu_1}{c}$
$N_2(s; \mu_2, \sigma_2) = \frac{1}{\sqrt{2\pi\sigma_2^2}} e^{-\frac{(s-\mu_2)^2}{2\sigma_2^2}}$
σ_2
$\frac{\sigma_1}{c}$
0
time of flight s

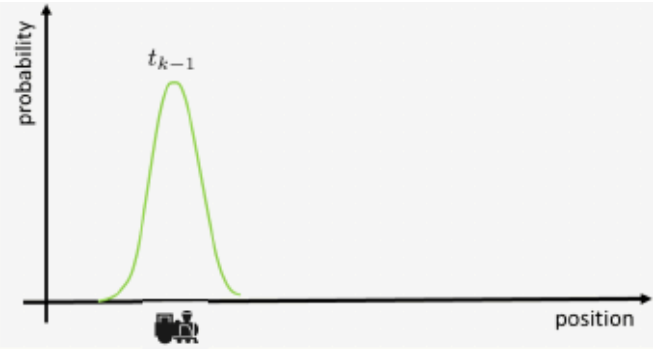




KF Algorithm Example



- No control input
- Assume constant velocity



$$\mathbf{x} = \begin{bmatrix} r \\ \dot{r} \end{bmatrix} \quad \mathbf{P} = \begin{bmatrix} \sigma_r^2 & \sigma_r \sigma_{\dot{r}} \\ \sigma_r \sigma_{\dot{r}} & \sigma_{\dot{r}}^2 \end{bmatrix}$$

Prediction

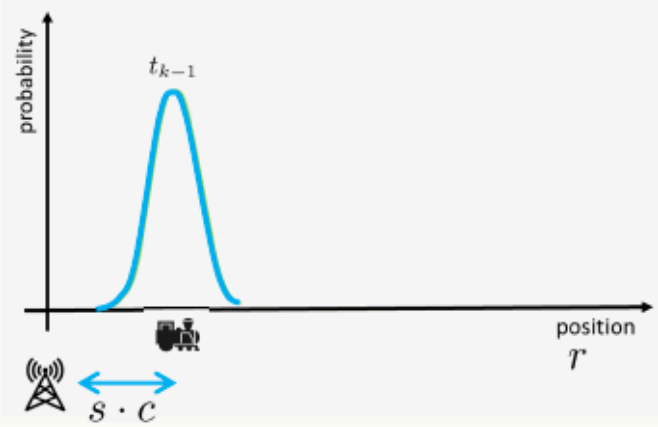
$$\begin{bmatrix} r \\ \dot{r} \end{bmatrix}_k^- = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ \dot{r} \end{bmatrix}_{k-1}$$

$$\mathbf{P}_k^- = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \mathbf{P}_{k-1} \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}_k^T + \overbrace{\begin{bmatrix} \sigma_{q1}^2 & 0 \\ 0 & \sigma_{q2}^2 \end{bmatrix}}^{\text{system noise } \mathbf{Q}}$$

Measurement

$$\mathbf{z} = \underbrace{\begin{bmatrix} \frac{1}{c} & 0 \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} r \\ \dot{r} \end{bmatrix} = \frac{r}{c} = s$$

$$\mathbf{R} = \sigma_s^2$$



Correction

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k (\mathbf{z}_k - \mathbf{H}_k \mathbf{x}_k^-) \\ \mathbf{P}_k &= (\mathbb{1} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- \end{aligned}$$

System Noise Q

$$\begin{bmatrix} r \\ \dot{r} \end{bmatrix}_k^- = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ \dot{r} \end{bmatrix}_{k-1}$$

$$r_k = r_{k-1} + \dot{r}_{k-1} \cdot \Delta t$$

$$\dot{r}_k = \dot{r}_{k-1} + a_{noise} \Delta t$$

model uncertainty in velocity update as **acceleration noise**

$$N(a_{noise}; 0, \sigma_a)$$

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_a^2 \Delta t \end{bmatrix}$$